Assignment 3

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Part 1

After implementing PCA, we could see what proportion of the variance each component explained. By summing the components in descending order of proportion size, the number of components needed to obtain 95% of the total variance was 35.

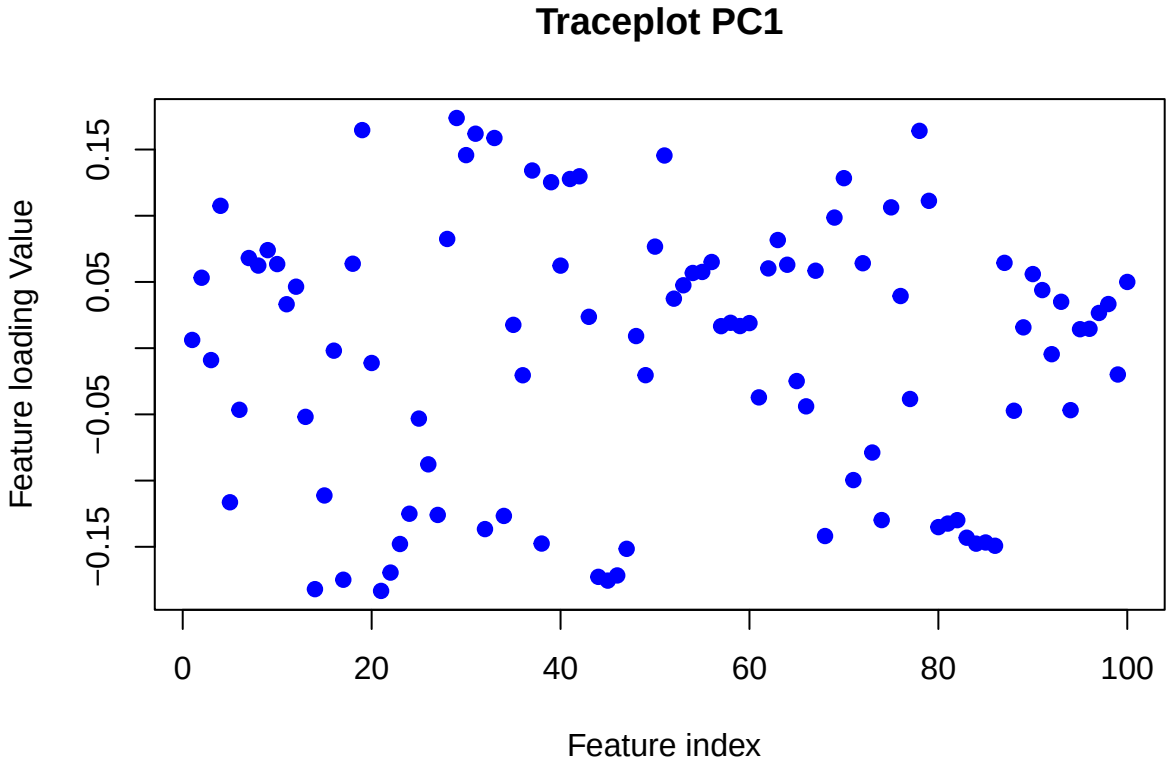
The proportion of variation explained by each of the first two principal components were:

PC1 = **25.0169875**%

PC2 = **16.9359743**%

Part 2

Many features have a notable contribution to this component, but the values are scattered and cover the entire range from around -0.18 to 0.17 which suggests that the component is not dominated by just one or two features.

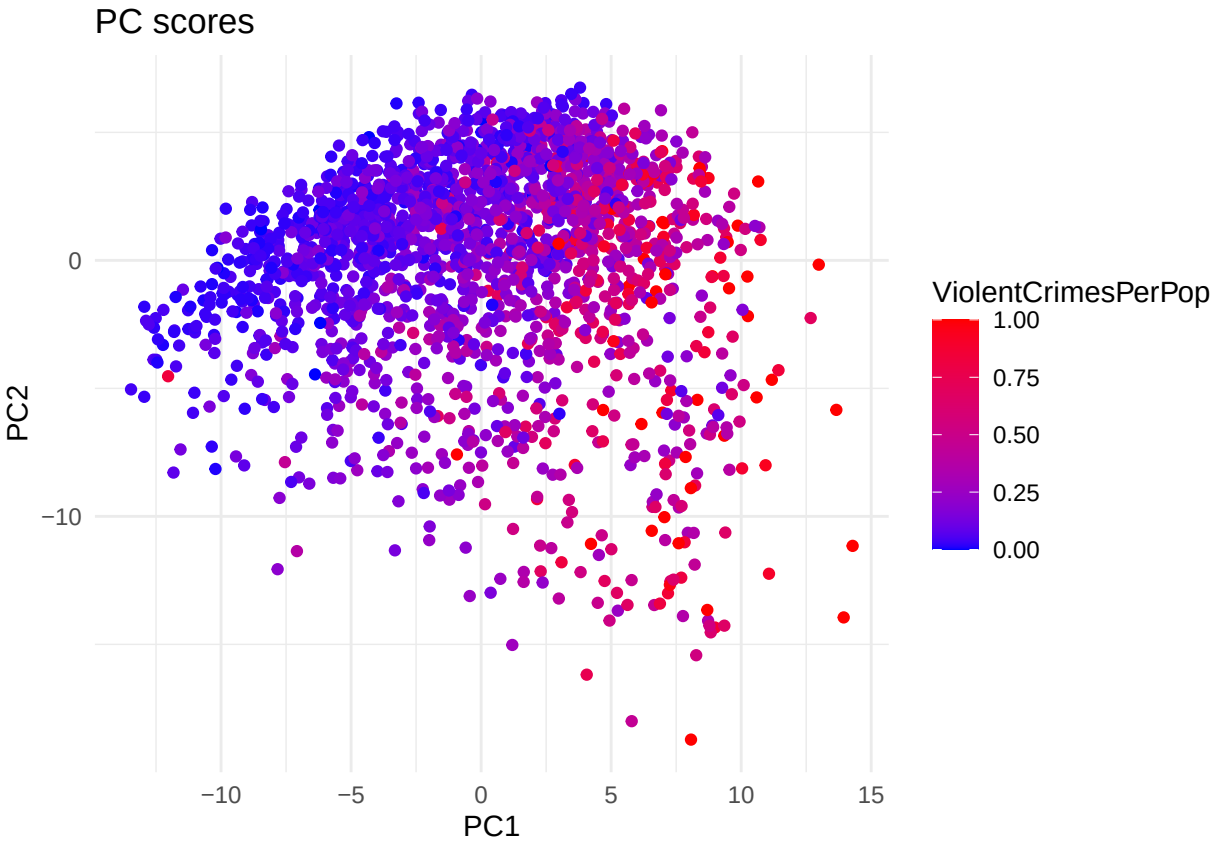


Below is a table of the most contributing factors to PC1.

Table 1: Top 5 Contributing Features to PC1

Feature	Loading
medFamInc	0.1833080
medIncome	0.1819830
PctKids2Par	0.1755423
pctWInvInc	0.1748683
PctPopUnderPov	0.1737978

We can clearly see the relationship between the most contributed features relating to crime as the component captures the lack of economic resources and social stability that links directly to elevated levels of violent crime.



Above is the plot of PC scores in the coordinates (PC1, PC2). On the x axis, PC1, which is the primary dimension and accounts for the largest amount of variance, there is a clear gradient of color from left to right. This means the linear combination of features that define PC1 (including the top 5 features mentioned earlier) are good predictors for violent crime rate.

The y axis, PC2, accounts for the second largest amount of variance. The spread is quite big, but the color does not change as clearly with vertical movement. For example, at PC1=0, you see a mix of colors across the vertical range. This means that PC2 captures variation in the data but this variation is not as strongly related to the violent crime rate as PC1.

Part 3

We scaled the data with a 50/50 ratio and fitted a linear regression model.

Calculated mean squared error (MSE):

$\text{MSE}_{\text{train}}$ = **0.2752071**

MSE_{test} = **0.4248011**

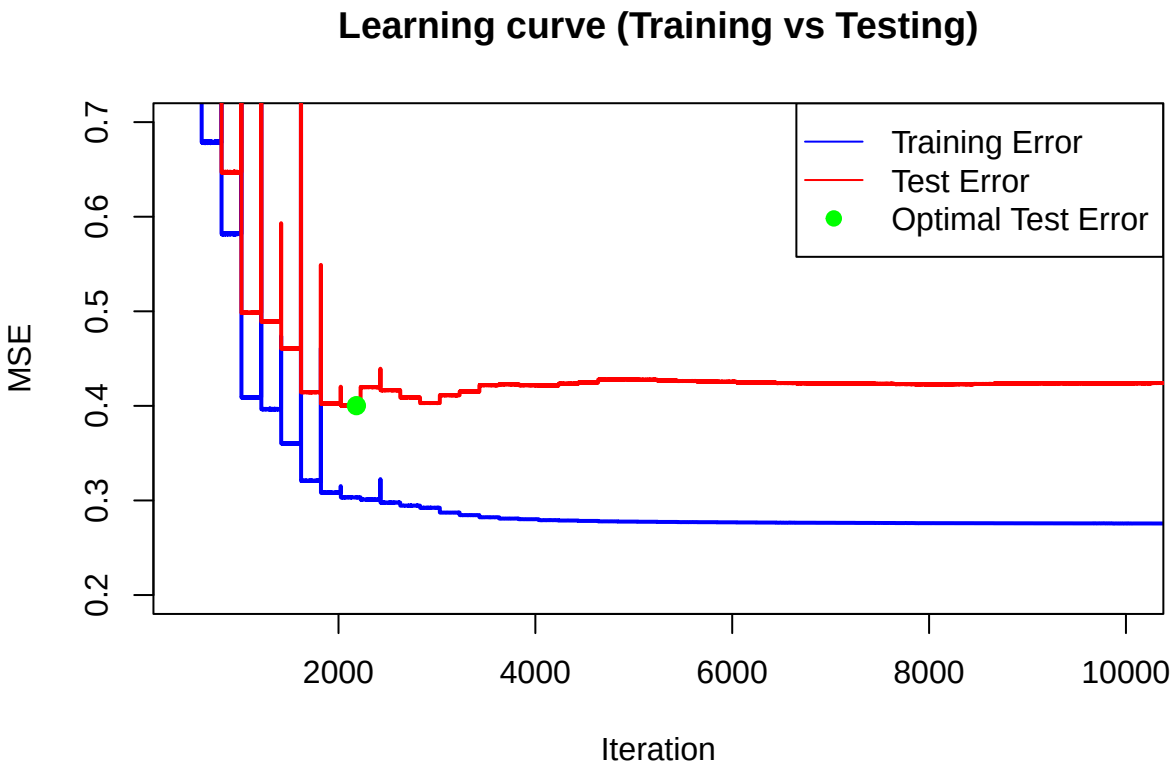
The training MSE is relatively low, but the test MSE is significantly higher ($0.4248/0.2752 \approx 1.54$). The test error is about 54% higher than the training error. This is a case of overfitting as there is a significant gap between the training and test MSE, indicating that the model has captured random noice in the training data. The quality of the model is therefore suboptimal because it is to complex for the amount of data provided, leading to high variance and poor performance on unseen data.

Part 4

We implemented MSE as the cost function that depends on theta as this is typically the cost function used in linear regression. Here, y_i is the i:th actual y value and the expression $\theta^T x_i$ is the i:th predicted value $\hat{y}_i$.

$$\text{Cost function} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \theta^T x_i)^2$$

Below is the plot showing dependence of both errors on the iteration number.



According to the early stopping criterion, iteration **2182** is optimal. In the optimal model, these are the computed training and test errors:

Optimal-model-$\text{MSE}_{\text{train}}$ = **0.3032999**

Optimal-model-MSE_{test} = **0.4002329**

The optimal model found using early stopping has a lower test MSE, but higher training MSE compared to the lm() model in part 3, even though the cost function used was the same. By stopping the optim() early at the lowest test MSE we avoided the phase where the model started to overfit, as can be seen in the graph

above. Early stopping produced a more robust set of parameters θ that generalize (a little bit) better to the test set.